
SYSTEM ANALYSIS DOCUMENT (SAD)

UMHackathon 2026

PeopleGraph

ChatGPT 2.0

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1. Introduction

PeopleGraph is an AI-powered Decision Support System (DSS) HR platform purpose-built for Malaysian Small and Medium Enterprises (SMEs). It introduces a strategic solution to transform traditional, error-prone, spreadsheet-based HR management into an intelligent, data-driven platform that automates statutory compliance, detects workforce anomalies, and provides predictive analytics.

1.1 Purpose:

This system analysis document highlights the technical scope, architectural design, and strategic decisions behind the development of PeopleGraph. It serves as a blueprint for the system's capabilities, data flows, and integration of AI services.

The key elements of the system that have been covered are as follows:

- Architecture:

A client-server architecture deployed with a FastAPI backend and a Next.js frontend, utilizing a microservice-inspired modular design within a monolithic repository.

- Data Flows:

How HR data (attendance, payroll, performance) flows from input, moves through the FastAPI backend, is processed by AI engines (Isolation Forest, Ensemble Forecasting, LLMs), and is stored in PostgreSQL (with pgvector) before returning insights to the frontend dashboard.

- Model Process:

The end-to-end workflow of core features: running payroll with data fallbacks, anomaly detection, predictive hiring impact, and the new Cuti Peristiwa Shock simulation.

- Role of Reference:

This document serves as a high-level reference for developers, testers, and stakeholders mapping the technical boundaries of the project.

1.2 Background:

Malaysian SMEs face critical challenges in HR administration. Most rely on manual Excel spreadsheets, leading to compliance errors with complex statutory bodies (EPF, SOCSO, EIS, PCB) and hidden workforce inefficiencies.

1.2.1 Previous Version:

The existing "system" for most target users consists of disjointed Excel spreadsheets, manual punch cards, and standalone LHDN calculator websites. Data is siloed, making real-time analytics impossible.

1.2.2 Changes in Major Architectural Components and New Capabilities:

New capabilities introduced in PeopleGraph include:

- Automated Statutory Calculator Engine with Data Fallback Logic.
- AI DSS Layer: Isolation Forest for Multivariate Anomaly Detection and Ensemble Forecasting (scikit-learn) for Revenue Prediction.
- NLP & Generative Processor: YTL AI Labs ILMU-GLM-5.1 for Manglish NLP Sentiment Analysis and Semantic SaaS Onboarding.
- Digital Twin & Risk Simulation: Visual workforce restructuring (D3.js) and the Cuti Peristiwa Shock simulation.

1.3 Target Stakeholders:

Stakeholders	Roles	Expectations
SME Owner	View dashboards, run what-if simulations, review anomalies	Clear visibility into labor costs, actionable AI-driven insights for restructuring, and evaluating public holiday operational risks.
HR Admin	Run payroll, export compliance reports, manage employees	Automated EPF/SOCSO calculations, strict LHDN name formatting, and easy batch processing with automated fallbacks for missing data.
Employee	Participate in pulse surveys, onboarding	Anonymous feedback mechanisms and tailored SaaS package suggestions via AI.
Development Team	Build and maintain the platform	Clear API contracts, modular architecture, and extensible AI service integrations.
QA Team	Validate statutory math and system behavior	Defined edge cases for compliance rules, RBAC security testing, and fallback data verification.

2. System Architecture & Design

2.1 High Level Architecture:

2.1.1 Overview:

Type	Details
System	Web Application (Responsive Dashboard)
Architecture	Client-Server (Next.js Frontend + FastAPI Backend)

PeopleGraph is structured as a client-server system. The client is a Next.js React application used by SME Owners and HR Admins. It communicates via RESTful APIs with the FastAPI backend. The backend is logically divided into modular routers (Compliance, Dashboard, AI Analytics, Employees) and interacts with a PostgreSQL database via SQLAlchemy ORM.

2.1.2 LLM as Service Layer:

The architecture integrates YTL AI Labs ILMU-GLM-5.1 as a specialized service layer for:

- **Semantic Onboarding:**
Analyzing free-text job descriptions to recommend customized software/SaaS packages.
- **Sentiment Analysis:**
Processing unstructured Manglish feedback to categorize team energy levels.

2.1.3 Dependency & Integration Workflow:

- **Prompt Construction:**
The backend service constructs specialized prompts, injecting raw text and instructing the model to return strict JSON schema.
- **Context Window:**
Includes system instructions, the specific task, and the user's raw input.
- **Parsing Responses:**
The FastAPI backend receives the response, parses the JSON, and maps it to internal data schemas (SimulationResponse or TeamEnergyData).
- **Major API Calls:**
 - Frontend → POST /api/v1/ai/onboarding/suggest-package

- Backend → External POST <https://api.zai.ytl.com/v1/chat/completions> (ILMU-GLM-5.1)
- Backend → PostgreSQL

2.1.4 Sequence Diagram

1. HR Admin initiates "Run Payroll" on the Frontend Dashboard.
2. Frontend sends POST /api/v1/payroll/runs.
3. FastAPI Backend queries the Database for active Employee records.
4. Backend calls StatutoryCalculator to compute EPF, SOCSO, EIS, PCB.
5. HR Admin requests Compliance Export POST /api/v1/payroll/compliance/export.
6. ComplianceExporter applies strict LHDN name formatting and invokes data fallback logic if columns are missing.
7. Backend saves records and returns the formatted payload.
8. (Conceptual) Backend calculates the SHA-256 hash and sends it to the BlockchainManager (Polygon L2 Escrow Ledger).

2.2 Technological Stack:

- Frontend: Next.js 16, React 19, Tailwind CSS 4, D3.js.
- Backend: FastAPI (Python 3.10+), SQLAlchemy (Async), scikit-learn, pandas.
- Database: PostgreSQL 14+ with pgvector (production) / SQLite (local dev).
- Deployment: Docker containerization.
- External APIs: YTL AI Labs ILMU-GLM-5.1.

2.3 Key Data Flows:

2.3.1 Data Flow Diagram (Conceptual):

- **Process 1 (Statutory):**
Salary data → Data Fallback Logic → Statutory Calculator → Compliance Exporter (Strict Formatting).
- **Process 2 (Predictive AI):**
Proposed Hire Input → Ensemble Forecasting Model → Revenue Impact vs. Automation Potential mapping.
- **Process 3 (Simulation):**
Operational Capacity + Holiday Stats → Cuti Peristiwa Shock Engine → SLA impact vs. Wage Penalty.
- **Process 4 (NLP):**
Unstructured Manglish feedback / Role Descriptions → ILMU-GLM-5.1 API → Structured Output (Sentiment/Packages).
- **Storage:**
Aggregated metrics are served via a unified /api/v1/dashboard/overview endpoint to minimize frontend round-trips.

2.3.2 Normalized Database Schema (Highlights):

Designed to 3NF to ensure data integrity.

- User (1:1) Role (RBAC permissions)
- Department (1:N) Employee
- Employee (1:N) PerformanceSnapshot (Anomaly flags)
- PayrollRun (1:N) PayrollLineItem
- PayrollLineItem (N:1) Employee

3.. Functional Requirements & Scope

3.1 Minimum Viable Product (MVP)

Features	Description
Statutory Payroll Engine	Automated calculation of EPF, SOCSO, EIS, and PCB with strict LHDN name formatting and missing data fallback logic.
Anomaly Detection	ML-based (Isolation Forest) detection of multivariate attendance and productivity anomalies.
Cuti Peristiwa Simulation	Predictive simulation calculating the trade-off between statutory wage penalties and warehouse SLA impact during sudden holidays.
Hiring Prediction & Onboarding	Predicts revenue impact of new hires and uses ILMU-GLM-5.1 to suggest SaaS onboarding packages.
Dashboard Overview	Single-pane unified API endpoint serving Finance Cards, 30-day Attendance Heatmaps, Team Energy, and concentric Department splits.

3.2 Non-Functional Requirements (NFRs)

Quality	Requirements	Implementation
Security (RBAC)	Strict access control to financial data.	Granular dependencies require <code>payroll_access</code> in FastAPI routes.
Compliance Accuracy	Must adhere strictly to Malaysian law.	Dedicated StatutoryCalculator and automated fallback logging in ComplianceExporter.
Performance	Dashboard must load quickly despite complex aggregations.	Unified /overview endpoint fetching all metrics in parallel asynchronously.
Token Latency	LLM API calls must not block the main thread.	Use of asynchronous HTTP clients in FastAPI for external calls.

3.3 Out of Scope / Future Enhancement

- Full production-level blockchain deployment (Escrow Ledger)
- Mobile application (iOS/Android)
- Multi-tenant enterprise architecture
- Advanced leave management system
- Real-time external API integrations (e.g., LHDN, direct bank API)

4. Monitor, Evaluation, Assumptions & Dependencies

4.1 Technical Evaluation:

- **RBAC Auditing:**

Ensure API endpoints strictly enforce permissions.

- **Fallback Verification:**

Validate that the ComplianceExporter correctly identifies missing data (0 or blank) and seamlessly applies fallback values while logging the event.

4.2 Priority Matrix:

- **P1 Critical:**

Incorrect statutory calculations or data leak across RBAC roles. Action: Immediate hotfix and database audit.

- **P2 High:**

ILMU-GLM-5.1 API unavailable. Action: Fallback to heuristic mocking for sentiment and onboarding to ensure application availability.

4.3 Monitoring:

- Logging of data fallback triggers in ComplianceExporter to identify systemic data entry issues by HR.
- Log monitoring for Isolation Forest cold-starts.

4.4 Assumptions:

- Malaysian minimum wage remains RM1,700 for baseline calculations.
- SMEs have sufficient historical data (at least 20 records) for the Ensemble Forecasting model to train accurately.
- Target users will primarily use modern web browsers compatible with Next.js 16 and D3.js.

4.5 External Dependencies:

Tools	Purpose	Risks
YTL AI Labs ILMU-GLM-5.1	Core LLM engine for Manglish sentiment and semantic onboarding.	High - API rate limits or downtime. Mitigation: Implement heuristic fallback logic.
Polygon L2	Immutable audit trail for payroll records.	Low - Simulated/mockred in current hackathon phase.

5. Project Management & Team Contributions

5.1 Project Timeline:

- **Day 1-2:**
Conceptualization, Schema Design, and UI Mockups.
- **Day 3-5:**
Backend setup, Statutory Calculator logic, and API Routing.
- **Day 6-7:**
Frontend integration, D3.js Visualizations, and AI Model integration (Simulation & Anomaly).
- **Day 8-10:**
Testing, documentation, and refinement of NLP prompts.

5.2 Team Members Role:

Name	Role
Teo Zheng Xi	Back End Developer
Chong Pohyi	Front End Developer
Teoh Xi Xian	Front End Developer
Tan Wei Feng	Documentation
Lim Jie Shin	Documentation

5.3 Recommendation:

- Implement Redis caching for Dashboard analytics to further reduce PostgreSQL load during peak usage.
- Transition anomaly detection and heavy predictive simulations to an asynchronous task queue (e.g., Celery) for background processing on larger datasets.